

1. Write a Pandas program to create a scatter plot of the trading volume/stock prices of Alphabet Inc. stock between two specific dates.  
**alphabet\_stock\_data:**

| Date       | Open     | High     | Low      | Close   | Adj Close | Volume  |
|------------|----------|----------|----------|---------|-----------|---------|
| 01-04-2020 | 1122     | 1129.69  | 1097.45  | 1105.62 | 1105.62   | 2343100 |
| 02-04-2020 | 1098.26  | 1126.86  | 1096.4   | 1120.84 | 1120.84   | 1964900 |
| 03-04-2020 | 1119.015 | 1123.54  | 1079.81  | 1097.88 | 1097.88   | 2313400 |
| 06-04-2020 | 1138     | 1194.66  | 1130.94  | 1186.92 | 1186.92   | 2664700 |
| 07-04-2020 | 1221     | 1225     | 1182.23  | 1186.51 | 1186.51   | 2387300 |
| 08-04-2020 | 1206.5   | 1219.07  | 1188.16  | 1210.28 | 1210.28   | 1975100 |
| 09-04-2020 | 1224.08  | 1225.57  | 1196.735 | 1211.45 | 1211.45   | 2175400 |
| 13-04-2020 | 1209.18  | 1220.51  | 1187.598 | 1217.56 | 1217.56   | 1739800 |
| 14-04-2020 | 1245.09  | 1282.07  | 1236.93  | 1269.23 | 1269.23   | 2470400 |
| 15-04-2020 | 1245.61  | 1280.46  | 1240.4   | 1262.47 | 1262.47   | 1671700 |
| 16-04-2020 | 1274.1   | 1279     | 1242.62  | 1263.47 | 1263.47   | 2518100 |
| 17-04-2020 | 1284.85  | 1294.43  | 1271.23  | 1283.25 | 1283.25   | 1949000 |
| 20-04-2020 | 1271     | 1281.6   | 1261.37  | 1266.61 | 1266.61   | 1695500 |
| 21-04-2020 | 1247     | 1254.27  | 1209.71  | 1216.34 | 1216.34   | 2153000 |
| 22-04-2020 | 1245.54  | 1285.613 | 1242     | 1263.21 | 1263.21   | 2093100 |
| 23-04-2020 | 1271.55  | 1293.31  | 1265.67  | 1276.31 | 1276.31   | 1566200 |
| 24-04-2020 | 1261.17  | 1280.4   | 1249.45  | 1279.31 | 1279.31   | 1640400 |
| 27-04-2020 | 1296     | 1296.15  | 1269     | 1275.88 | 1275.88   | 1600600 |
| 28-04-2020 | 1287.93  | 1288.05  | 1232.2   | 1233.67 | 1233.67   | 2951300 |
| 29-04-2020 | 1341.46  | 1359.99  | 1325.34  | 1341.48 | 1341.48   | 3793600 |
| 30-04-2020 | 1324.88  | 1352.82  | 1322.49  | 1348.66 | 1348.66   | 2665400 |
| 01-05-2020 | 1328.5   | 1352.07  | 1311     | 1320.61 | 1320.61   | 2072500 |

SOLUTION :

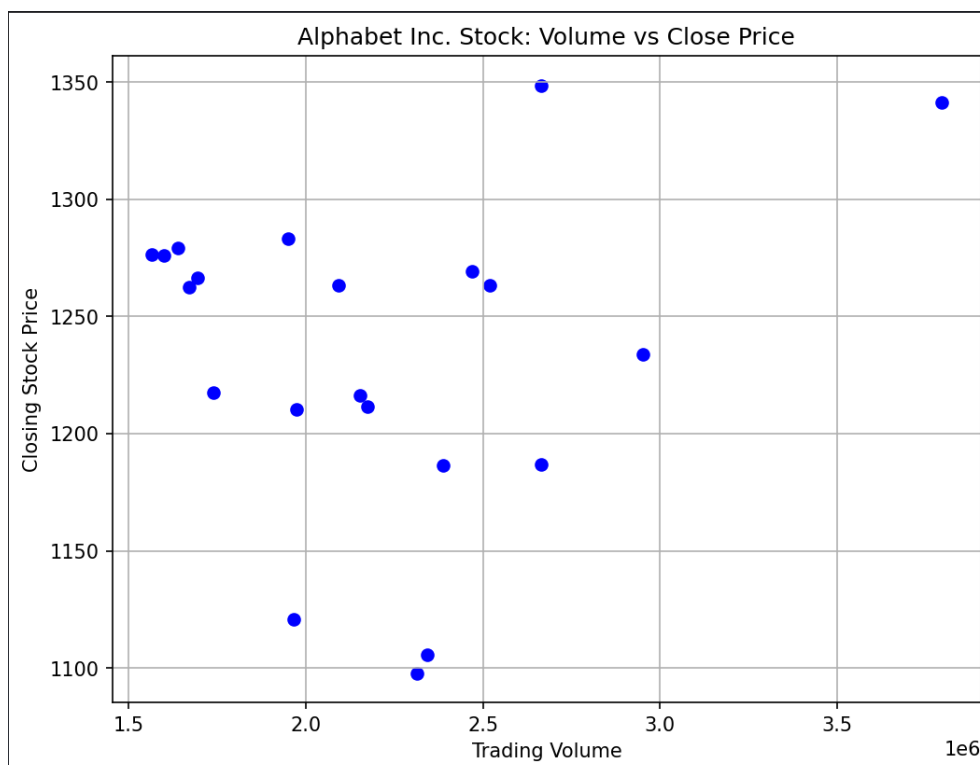
```

stock_plot.py X
C:\Users\> Vedvyass M > Music > Nutri-vision > stock_analysis > stock_plot.py
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Read the CSV file
5 df = pd.read_csv("alphabet_stock_data.csv")
6
7 # Convert Date column to datetime format
8 df['Date'] = pd.to_datetime(df['Date'], format='%d-%m-%Y')
9
10 # Select data between two dates
11 start_date = '2020-04-01'
12 end_date = '2020-04-30'
13
14 filtered_data = df[(df['Date'] >= start_date) & (df['Date'] <= end_date)]
15
16 # Scatter Plot
17 plt.figure(figsize=(8,6))
18 plt.scatter(filtered_data['Volume'], filtered_data['Close'], color='blue')
19
20 plt.title("Alphabet Inc. Stock: Volume vs Close Price")
21 plt.xlabel("Trading Volume")
22 plt.ylabel("Closing Stock Price")
23 plt.grid(True)
24
25 plt.savefig("scatter_plot.png", dpi=150, bbox_inches='tight')
26 print("Plot saved as scatter_plot.png")
27 print(f"\nFiltered data ({len(filtered_data)} rows):")
28 print(filtered_data[['Date', 'Volume', 'Close']].to_string(index=False))
29

```

|    | A          | B        | C        | D        | E       | F         | G       | H | I |
|----|------------|----------|----------|----------|---------|-----------|---------|---|---|
| 1  | Date       | Open     | High     | Low      | Close   | Adj Close | Volume  |   |   |
| 2  | 01-04-2020 | 1122     | 1129.69  | 1097.45  | 1105.62 | 1105.62   | 2343100 |   |   |
| 3  | 02-04-2020 | 1098.26  | 1126.86  | 1096.4   | 1120.84 | 1120.84   | 1964900 |   |   |
| 4  | 03-04-2020 | 1119.015 | 1123.54  | 1079.81  | 1097.88 | 1097.88   | 2313400 |   |   |
| 5  | 06-04-2020 | 1138     | 1194.66  | 1130.94  | 1186.92 | 1186.92   | 2664700 |   |   |
| 6  | 07-04-2020 | 1221     | 1225     | 1182.23  | 1186.51 | 1186.51   | 2387300 |   |   |
| 7  | 08-04-2020 | 1206.5   | 1219.07  | 1188.16  | 1210.28 | 1210.28   | 1975100 |   |   |
| 8  | 09-04-2020 | 1224.08  | 1225.57  | 1196.735 | 1211.45 | 1211.45   | 2175400 |   |   |
| 9  | 13-04-2020 | 1209.18  | 1220.51  | 1187.598 | 1217.56 | 1217.56   | 1739800 |   |   |
| 10 | 14-04-2020 | 1245.09  | 1282.07  | 1236.93  | 1269.23 | 1269.23   | 2470400 |   |   |
| 11 | 15-04-2020 | 1245.61  | 1280.46  | 1240.4   | 1262.47 | 1262.47   | 1671700 |   |   |
| 12 | 16-04-2020 | 1274.1   | 1279     | 1242.62  | 1263.47 | 1263.47   | 2518100 |   |   |
| 13 | 17-04-2020 | 1284.85  | 1294.43  | 1271.23  | 1283.25 | 1283.25   | 1949600 |   |   |
| 14 | 20-04-2020 | 1271     | 1281.6   | 1261.37  | 1266.61 | 1266.61   | 1695500 |   |   |
| 15 | 21-04-2020 | 1247     | 1254.27  | 1209.71  | 1216.34 | 1216.34   | 2153000 |   |   |
| 16 | 22-04-2020 | 1245.54  | 1285.613 | 1242     | 1263.21 | 1263.21   | 2093100 |   |   |
| 17 | 23-04-2020 | 1271.55  | 1293.31  | 1265.67  | 1276.31 | 1276.31   | 1566200 |   |   |
| 18 | 24-04-2020 | 1261.17  | 1280.4   | 1249.45  | 1279.31 | 1279.31   | 1640400 |   |   |
| 19 | 27-04-2020 | 1296     | 1296.15  | 1269     | 1275.88 | 1275.88   | 1600600 |   |   |
| 20 | 28-04-2020 | 1287.93  | 1288.05  | 1232.2   | 1233.67 | 1233.67   | 2951300 |   |   |
| 21 | 29-04-2020 | 1341.46  | 1359.99  | 1325.34  | 1341.48 | 1341.48   | 3793600 |   |   |
| 22 | 30-04-2020 | 1324.88  | 1352.82  | 1322.49  | 1348.66 | 1348.66   | 2665400 |   |   |
| 23 | 01-05-2020 | 1328.5   | 1352.07  | 1311     | 1320.61 | 1320.61   | 2072500 |   |   |

OUTPUT :



2 ) Write a Pandas program to create a Pivot table and find the maximum and minimum sale value of the items.(refer sales\_data table)

Sales\_data:

| OrderDate | Region  | Manager | SalesMan  | Item         | Units | Unit_price | Sale_amt    |
|-----------|---------|---------|-----------|--------------|-------|------------|-------------|
| 1-6-18    | East    | Martha  | Alexander | Television   | 95    | 1,198.00   | 1,13,810.00 |
| 1-23-18   | Central | Hermann | Shelli    | Home Theater | 50    | 500.00     | 25,000.00   |
| 2-9-18    | Central | Hermann | Luis      | Television   | 36    | 1,198.00   | 43,128.00   |
| 2-26-18   | Central | Timothy | David     | Cell Phone   | 27    | 225.00     | 6,075.00    |
| 3-15-18   | West    | Timothy | Stephen   | Television   | 56    | 1,198.00   | 67,088.00   |
| 4-1-18    | East    | Martha  | Alexander | Home Theater | 60    | 500.00     | 30,000.00   |
| 4-18-18   | Central | Martha  | Steven    | Television   | 75    | 1,198.00   | 89,850.00   |
| 5-5-18    | Central | Hermann | Luis      | Television   | 90    | 1,198.00   | 1,07,820.00 |
| 5-22-18   | West    | Douglas | Michael   | Television   | 32    | 1,198.00   | 38,336.00   |
| 6-8-18    | East    | Martha  | Alexander | Home Theater | 60    | 500.00     | 30,000.00   |
| 6-25-18   | Central | Hermann | Sigal     | Television   | 90    | 1,198.00   | 1,07,820.00 |
| 7-12-18   | East    | Martha  | Diana     | Home Theater | 29    | 500.00     | 14,500.00   |
| 7-29-18   | East    | Douglas | Karen     | Home Theater | 81    | 500.00     | 40,500.00   |
| 8-15-18   | East    | Martha  | Alexander | Television   | 35    | 1,198.00   | 41,930.00   |
| 9-1-18    | Central | Douglas | John      | Desk         | 2     | 125.00     | 250.00      |
| 9-18-18   | East    | Martha  | Alexander | Video Games  | 16    | 58.50      | 936.00      |
| 10-5-18   | Central | Hermann | Sigal     | Home Theater | 28    | 500.00     | 14,000.00   |
| 10-22-18  | East    | Martha  | Alexander | Cell Phone   | 64    | 225.00     | 14,400.00   |

SOLUTION :

```

File Edit Selection View Go Run Terminal Help
pivot_table.py X
C:\Users\Vedvyass M > Music > Nutri-vision > stock_analysis > pivot_table.py > ...
1 import pandas as pd
2
3 # Read the CSV file
4 df = pd.read_csv("sales_data.csv")
5
6 print("-" * 60)
7 print("          SALES DATA TABLE")
8 print("-" * 60)
9 print(df.to_string(index=False))
10
11 # Create a Pivot Table with Item as index and Sale_amt as values
12 # Aggregating with sum, max, and min
13 pivot_table = pd.pivot_table(
14     df,
15     index='Item',
16     values='Sale_amt',
17     aggfunc=['sum', 'max', 'min']
18 )
19
20 print("\n" + "-" * 60)
21 print("          PIVOT TABLE (Sale_amt by Item)")
22 print("-" * 60)
23 print(pivot_table)
24
25 # Find Maximum and Minimum sale value for each item
26 print("\n" + "-" * 60)
27 print("          MAXIMUM Sale Value of Each Item")
28 print("-" * 60)
29 max_sales = df.groupby('Item')['Sale_amt'].max()
30 print(max_sales.to_string())
31
32 print("\n" + "-" * 60)
33 print("          MINIMUM Sale Value of Each Item")
34 print("-" * 60)
35 min_sales = df.groupby('Item')['Sale_amt'].min()
36 print(min_sales.to_string())
37
38 # Overall max and min
39 print("\n" + "-" * 60)
40 print("          OVERALL MAX & MIN Sale Value")
41 print("-" * 60)
42 print(f"Maximum Sale Amount : {df['Sale_amt'].max():.2f} (Item: {df.loc[df['Sale_amt'].idxmax(), 'Item']})")
43 print(f"Minimum Sale Amount : {df['Sale_amt'].min():.2f} (Item: {df.loc[df['Sale_amt'].idxmin(), 'Item']})")
44

```

## OUTPUT :

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Vedvyass M\Music\Nutri-vision\stock_analysis> & 'C:\Users\Vedvyass M\AppData\Local\Programs\Python\Python38-64\python.exe' 'C:\Users\Vedvyass M\AppData\Local\Programs\Python\Python38-64\python\launcher' '53594' '---' 'C:\Users\Vedvyass M\Music\Nutri-vision\stock_analysis\pivot_table.py'
Video Games          936.0

=====
OVERALL MAX & MIN Sale Value
=====
Maximum Sale Amount : 113810.00 (Item: Television)
Minimum Sale Amount : 250.00 (Item: Desk)
PS C:\Users\Vedvyass M\Music\Nutri-vision\stock_analysis>
```